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Inventor(s) CHEN; Minghui et al.

EMBEDDED ELECTRONIC FUSES

Abstract

A system includes a read amplifier having a first input, a second input, a first output, and a second output. The read amplifier includes a first inverter having an input and an output, wherein the output of the first inverter is coupled to the first output of the read amplifier, and a second inverter having an input and an output, wherein the output of the second inverter is coupled to the second output of the read amplifier and the input of the first inverter, and the input of the second inverter is coupled to the output of the first inverter. The system also includes one or more first fuses coupled to the first input of the read amplifier, and one or more second fuses coupled to the second input of the read amplifier.

Inventors: **CHEN; Minghui (San Diego, CA), TANG; Yiwu (San Diego, CA)**

Applicant: **QUALCOMM Incorporated** (San Diego, CA)

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Background/Summary

BACKGROUND

Field

[0001] Aspects of the present disclosure relate generally to non-volatile memory, and, more particularly, to embedded electronic fuses.

Background

[0002] An electronic device may include non-volatile memory (e.g., for storing device settings and/or other information). The non-volatile memory may include electronic fuses (also referred to as E-fuse or eFuse) where each fuse may store a bit value of one or zero depending on whether the electronic fuse is blown or unblown.

SUMMARY

[0003] The following presents a simplified summary of one or more implementations in order to provide a basic understanding of such implementations. This summary is not an extensive overview of all contemplated implementations and is intended to neither identify key or critical elements of all implementations nor delineate the scope of any or all implementations. Its sole purpose is to present some concepts of one or more implementations in a simplified form as a prelude to the more detailed description that is presented later.

[0004] A first aspect relates to a system. The system includes a read amplifier having a first input, a second input, a first output, and a second output. The read amplifier includes a first inverter having an input and an output, wherein the output of the first inverter is coupled to the first output of the read amplifier, and a second inverter having an input and an output, wherein the output of the second inverter is coupled to the second output of the read amplifier and the input of the first inverter, and the input of the second inverter is coupled to the output of the first inverter. The system also includes one or more first fuses coupled to the first input of the read amplifier, and one or more second fuses coupled to the second input of the read amplifier.

[0005] A second aspect relates to a method for reading a bit stored in electronic fuses. The electronic fuses include one or more first fuses and one or more second fuses. The method includes sensing a resistance difference between the one or more first fuses and the one or more second fuses using a read amplifier, the read amplifier including a first inverter and a second inverter that are cross coupled, latching a logic value at an output of the first inverter to obtain a first latched logic value, latching a logic value at an output of the second inverter to obtain a second latched logic value, wherein the first latched logic value and the second latched logic value are complementary, and outputting at least one of the first latched logic value and the second latched logic value.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. **1** shows an example of a device including an energy harvester and a non-volatile memory according to certain aspects of the present disclosure.

[0007] FIG. **2** shows an example of a read amplifier configured to read a bit stored on electronic fuses according to certain aspects of the present disclosure.

[0008] FIG. **3** shows an exemplary implementation of inverters in the read amplifier of FIG. **2** according to certain aspects of the present disclosure.

[0009] FIG. **4** shows an example of a write circuit configured to write a bit value according to certain aspects of the present disclosure.

[0010] FIG. **5** shows an example of a read circuit according to certain aspects of the present disclosure.

[0011] FIG. **6** shows an example of the read circuit of FIG. **6** further including a gating circuit and an enable circuit according to certain aspects of the present disclosure.

[0012] FIG. **7** shows an exemplary implementation of a latch according to certain aspects of the present disclosure.
[0013] FIG. **8** shows an exemplary implementation of a done circuit according to certain aspects of the present disclosure.
[0014] FIG. **9** shows another example of a read circuit according to certain aspects of the present disclosure,
[0015] FIG. **10** shows an example of the read circuit of FIG. **9** further including a gating circuit and an enable circuit according to certain aspects of the present disclosure.
[0016] FIG. **11** shows an example of multiple read circuits coupled together according to certain aspects of the present disclosure.
[0017] FIG. **12** shows another example of multiple read circuits coupled together according to certain aspects of the present disclosure.
[0018] FIG. **13** is a flowchart illustrating an exemplary method for reading a bit stored in electronic fuses according to certain aspects of the present disclosure.

DETAILED DESCRIPTION

[0019] The detailed description set forth below, in connection with the appended drawings, is intended as a description of various configurations and is not intended to represent the only configurations in which the concepts described herein may be practiced. The detailed description includes specific details for the purpose of providing a thorough understanding of the various concepts. However, it will be apparent to those skilled in the art that these concepts may be practiced without these specific details. In some instances, well-known structures and components are shown in block diagram form in order to avoid obscuring such concepts.
[0020] FIG. **1** shows an example of an electronic device **100** according to certain aspects. The device **100** may include an Internet of Things (IoT) device, a sensor device, an asset tracker, a security device, a smart appliance, a wearable device, a communication device, a monitor, and so forth.
[0021] In the example in FIG. **1**, the device **100** includes an energy harvester **120** configured to convert ambient energy (i.e., energy from the surrounding environment) into electrical energy to power the device **100**. For example, the device **100** may be a battery-less device that relies on the energy harvester **120** for power. However, it is to be appreciated that the device **100** is not limited to a battery-less device and that the device **100** may also include a battery (not shown) in some implementations.
[0022] The energy harvester **120** may be implemented with any one of various types of energy harvesters. For example, the energy harvester **120** may be implemented with a radio receiver configured to convert radio frequency (RF) energy received by an antenna **125** into electrical energy. In another example, the energy harvester **120** may be implemented with a solar cell configured to convert solar energy into electrical energy. In still another example, the energy harvester **120** may be implemented with a piezoelectric transducer configured to convert vibrations or other types of mechanical energy into electrical energy. In yet another example, the energy harvester **120** may be implemented with a thermoelectric energy harvester configured to convert heat into electrical energy. In some implementations, the energy harvester **120** may include a combination of two or more different types of energy harvesters (e.g., a combination of two or more of the exemplary energy harvesters discussed above).
[0023] The device **100** also includes a non-volatile memory **135**, a storage capacitor **140**, a voltage monitor **145**, a power switch **155**, and a circuit **150**. Depending on the type of device, the circuit **150** may include one or more sensors (e.g., a temperature sensor, a humidity sensor, a biosensor, a gas sensor, a chemical sensor, a motion sensor, a proximity sensor, etc.), a wireless transmitter, and/or another type of circuit.
[0024] In this example, the energy harvester **120** is coupled to the storage capacitor **140** via a power bus **130** (also referred to as a supply bus or another type). The energy harvester **120** is

configured to output the electrical energy to the power bus **130**, and the storage capacitor **140** is configured to store the electrical energy in the form of electric charge. The electric charge on the storage capacitor **140** produces a voltage on the power bus **130** which is used to provide a supply voltage for the circuit **150**, as discussed further below.

[0025] In this example, the voltage monitor **145** is coupled to the power bus **130** and is configured to monitor the voltage on the power bus **130**. The power switch **155** is coupled between the power bus **130** and the circuit **150**. The on/off state of the power switch **155** is controlled by the voltage monitor **145**, which uses the power switch **155** to selectively couple the circuit **150** to the power bus **130** based on the monitored voltage on the power bus **130**. For example, the voltage monitor **145** may compare the monitored voltage on the power bus **130** with a voltage threshold, turn off the power switch **155** if the monitored voltage is below the voltage threshold, and turn on the power switch **155** if the monitored voltage is equal to or greater than the voltage threshold. Thus, in this example, the circuit **150** is coupled to the power bus **130** to receive power when the voltage on the power bus **130** reaches the voltage threshold.

[0026] In the example shown in FIG. **1**, the power switch **155** includes a p-type field effect transistor (PFET) **158**, in which the gate of the PFET **158** is coupled to the voltage monitor **145**. In this example, the voltage monitor **145** may output the voltage on the power bus **130** to the gate of the PFET **158** to turn off the power switch **155**, and couple the gate of the PFET **158** to ground potential to turn on the PFET **158**. However, it is to be appreciated that the power switch **155** is not limited to this example, and that the power switch **155** may be implemented with other types of switches.

[0027] The non-volatile memory **135** is coupled to the power bus **130** to receive power. The non-volatile memory **135** may be used to store one or more settings for the voltage monitor **145** and/or the energy harvester **120**. For example, the non-volatile memory **135** may store the voltage threshold for the voltage monitor **145**. In some implementations, the voltage threshold may be a function of the process corner of the circuit **150**. For example, the voltage threshold may be higher for a slow-slow (SS) process corner and lower for a fast-fast (FF) process corner. In this example, the non-volatile memory **135** may include process information for the circuit **150**. For the example where the energy harvester **120** includes a receiver for harvesting RF energy, the non-volatile memory may store frequency tuning information for the receiver.

[0028] The non-volatile memory **135** may be implemented with electronic fuses (also referred to as E-fuse or eFuse), in which each electronic fuse may store a bit value of one or zero depending on whether the electronic fuse is blown or unblown. For example, an electronic fuse may be left unblown to store a bit value of zero, and blown to store a bit value of one. The resistance of an unblown fuse may be low (e.g., less than 20 ohms) and the resistance of a blown fuse may be high (e.g., on the order of kilohms). Thus, the bit value stored by an electronic fuse may be read by sensing the resistance of the fuse.

[0029] To read the bit value of an electronic fuse, a read circuit may pre-charge a sense node high and couple the sense node to the electronic fuse. If the fuse is blown (i.e., the resistance of the fuse is high), then the sense node stays high. If the fuse is unblown (i.e., the resistance of the fuse is low), then the sense node is discharged through the fuse and goes low. The read circuit then determines whether the bit value of the fuse is one or zero based on whether the sense node is high or low.

[0030] The read circuit discussed above may require a supply voltage of at least 0.8 volts to operate and a clock signal (e.g., 1 MHz to 10 MHz clock signal) for timing operations of the read circuit. In addition, the read circuit may consume a relatively large amount of current to sense whether an electronic fuse is blown or unblown (e.g., current on the order of milliamps to sense an unblown fuse).

[0031] A challenge with implementing the non-volatile memory **135** using electronic fuses is that the energy harvester **120** provides a small amount of energy, which may not be sufficient to power the read circuit discussed above. For example, the energy collected by the energy harvester **120** may be well below the amount of energy needed to generate the clock signal for the read circuit and provide the relatively large current consumed by the read circuit for sensing operations. Further, the voltage on the power bus **130** may be much lower than the minimum voltage required for the read circuit to operate.

[0032] FIG. **2** shows an example of a read amplifier **210** according to certain aspects of the present disclosure. The read amplifier **210** uses positive feedback to sense a resistance difference between one or more blown fuses and one or more unblown fuses. As discussed further below, this allows the read amplifier **210** to read a bit using a low voltage and low current, making the read amplifier **210** suitable for use in a device (e.g., the device **100**) powered by an energy harvester (e.g., the energy harvester **120**).

[0033] The read amplifier **210** has a first input **212**, a second input **214**, a first output **216**, and a second output **218**. The first input **212** is coupled to one or more first fuses **250-1** to **250-2**, and the second input **214** is coupled to one or more second fuses **260-1** to **260-2**. In the example shown in FIG. **2**, the one or more first fuses **250-1** to **250-2** are coupled in series between the first input **212** of the read amplifier **210** and ground, and the one or more second fuses **260-1** to **260-2** are coupled in series between the second input **214** of the read amplifier **210** and ground. FIG. **2** shows an example where the one or more first fuses **250-1** to **250-2** include two fuses. However, it is to be appreciated that the one or more first fuses **250-1** to **250-2** may include a single fuse or two or more fuses (e.g., four fuses, eight fuses, sixteen fuses, etc.) in other examples. FIG. **2** also shows an example where the one or more second fuses **260-1** to **260-2** include two fuses. However, it is to be appreciated that the one or more second fuses **260-1** to **260-2** may include a single fuse or two or more fuses (e.g., four fuses, eight fuses, sixteen fuses, etc.) in other examples.

[0034] In this example, the one or more first fuses **250-1** to **250-2** and the one or more second fuses **260-1** to **260-2** are used to store a bit value. For example, to store a bit value of zero, the one or more first fuses **250-1** to **250-2** may be unblown and the one or more second fuses **260-1** to **260-2** may be blown. To store a bit value of one, the one or more first fuses **250-1** to **250-2** may be blown and the one or more second fuses **260-1** to **260-2** may be unblown. Exemplary circuitry for blowing the one or more first fuses **250-1** to **250-2** or the one or more second fuses **260-1** to **260-2** is discussed below with reference to FIG. **4**.

[0035] Thus, when a bit value of zero or one is stored, one of the inputs **212** and **214** of the read amplifier **210** is coupled to one or more blown fuses and the other one of the inputs **212** and **214** of the read amplifier **210** is coupled to one or more unblown fuses. This creates a large resistance difference between the inputs **212** and **214** of the read amplifier **210** in which the resistance of the input coupled to the one or more blown fuses is much higher than the resistance of the input coupled to the one or more unblown fuses. The resistance difference between the inputs **212** and **214** of the read amplifier **210** may be increased by increasing the number of the one or more first fuses **250-1** and **250-2** and increasing the number of the one or more second fuses **260-1** and **260-2**. As discussed further below, the read amplifier **210** reads the stored bit value by sensing the resistance difference between the inputs **212** and **214**.

[0036] In this example, the read amplifier **210** includes a first inverter **220** and a second inverter **230** that are cross coupled to provide positive feedback. As used herein, first and second inverters are cross coupled when the input of the first inverter is coupled to the output of the second inverter, and the input of the second inverter is coupled to the output of the first inverter. In this regard, the input **222** of the first inverter **220** is coupled to the output **234** of the second inverter **230**, and the input **232** of the second inverter **230** is coupled to the output **224** of the first inverter **220**. In this example, the output **224** of the first inverter **220** is coupled to the first output **216** of the read amplifier **210**, and the output **234** of the second inverter **230** is coupled to the second output **218** of the read amplifier **210**.

[0037] The first inverter **220** has a first terminal **226** coupled to a supply rail Vdd and a second terminal **228** coupled to the first input **212** of the read amplifier **210**. The first terminal **226** may also be referred to as a supply terminal or a power terminal. The first inverter **220** may be configured to pull the output **224** of the first inverter **220** to the potential at the first terminal **226** (e.g., Vdd) when the first inverter **220** is driven high and pull the output **224** of the first inverter **220** to the potential at the second terminal **228** when the first inverter **220** is driven low.

[0038] The second inverter **230** has a first terminal **236** coupled to the supply rail Vdd and a second terminal **238** coupled to the second input **214** of the read amplifier **210**. The second inverter **230** may be configured to pull the output **234** of the second inverter **230** to the potential at the first terminal **236** (e.g., Vdd) when the second inverter **230** is driven high and pull the output **234** of the second inverter **230** to the potential at the second terminal **238** when the second inverter **230** is driven low.

[0039] During operation, the positive feedback of the cross-coupled inverters **220** and **230** causes the read amplifier **210** to go into a first state or a second state based on the resistance difference at the inputs **212** and **214**. In the first state, the first output **216** is low (i.e., zero) and the second output **218** is high (i.e., one). In the second state, the first output **216** is high (i.e., one) and the second output **218** is low (i.e., zero). In this example, the first state may indicate a read bit value of zero, and the second state may indicate a read bit value of one, or vice versa.

[0040] When the one or more first fuses **250-1** to **250-2** are unblown and the one or more second fuses **260-1** to **260-2** are blown (e.g., stored bit value of zero), the resistance at the first input **212** of the read amplifier **210** is much lower than the resistance at the second input **214** of the read amplifier **210**. This resistance difference causes the positive feedback of the cross-coupled inverters **220** and **230** to push the read amplifier **210** into the first state in which the first output **216** is low (i.e., zero) and the second output **218** is high (i.e., one) indicating a read bit value of zero.

[0041] When the one or more first fuses **250-1** to **250-2** are blown and the one or more second fuses **260-1** to **260-2** are unblown (e.g., stored bit value of one), the resistance at the first input **212** of the read amplifier **210** is much higher than the resistance at the second input **214** of the read amplifier **210**. This resistance difference causes the positive feedback of the cross-coupled inverters **220** and **230** to push the read amplifier **210** into the second state in which the first output **216** is high (i.e., one) and the second output **218** is low (i.e., zero) indicating a read bit value of one.

[0042] In this example, the positive feedback of the cross-coupled inverters **220** and **230** allows the read amplifier **210** to sense the resistance difference between the inputs **212** and **214** of the read amplifier **210** (and hence read the stored bit value) using a low voltage and low current. This is because the positive feedback is able to push the read amplifier **210** into one of the first and second states indicating the read bit value based on the resistance difference using a low voltage and low current. As discussed above, the resistance difference between the inputs **212** and **214** of the read amplifier **210** is generated by blowing the one or more fuses on one side of the read amplifier **210** and leaving the one or more fuses on the other side of the read amplifier **210** unblown.

[0043] FIG. **3** shows an example in which each of the first inverter **220** and the second inverter **230** is implemented with a complementary inverter.

[0044] In this example, the first inverter **220** includes a first n-type field effect transistor (NFET) **310** and a first p-type field effect transistor (PFET) **320**. The drain of the first NFET **310** is coupled to the output **224**, the gate of the first NFET **310** is coupled to the input **222**, and the source of the first NFET **310** is coupled to the second terminal **228**. The drain of the first PFET **320** is coupled to the output **224**, the gate of the first PFET **320** is coupled to the input **222**, and the source of the first PFET **320** is coupled to the first terminal **226**. However, it is to be appreciated that the first inverter **220** is not limited to the exemplary implementation shown in FIG. **3**.

[0045] The second inverter **230** includes a second NFET **330** and a second PFET **340**. The drain of the second NFET **330** is coupled to the output **234**, the gate of the second NFET **330** is coupled to the input **232**, and the source of the second NFET **330** is coupled to the second terminal **238**. The drain of the second PFET **340** is coupled to the output **234**, the gate of the second PFET **340** is coupled to the input **232**, and the source of the second PFET **340** is coupled to the first terminal **236**. However, it is to be appreciated that the second inverter **230** is not limited to the exemplary implementation shown in FIG. **3**.

[0046] FIG. **4** shows an example of a write circuit **405** for programming the one or more first fuses **250-1** and **250-2** and the one or more second fuses **260-1** and **260-2** with a bit value. The write circuit **405** includes a blow driver **450** configured to blow one or more fuses by driving high current through the one or more fuses, as discussed further below. The write circuit **405** also includes a blow selector **460** configured to select the one or more first fuses **250-1** and **250-2** or the one or more second fuses **260-1** and **260-2** for a blow operation by the blow driver **450** depending on the bit value being stored.

[0047] The write circuit **405** also includes transistors **410-1** and **410-2** and transistors **420-1** and **420-2** for selectively blowing the one or more first fuses **250-1** and **250-2**. Each of the transistors **410-1** and **410-2** is coupled between the blow driver **450** and a respective one of the one or more first fuses **250-1** and **250-2**, and each of the transistors **420-1** and **420-2** is coupled between a respective one of the one or more first fuses **250-1** and **250-2** and ground. The gates of the transistors **410-1**, **410-2**, **420-1**, and **420-2** are coupled to the blow selector **460**, which controls the on/off states of the transistors **410-1**, **410-2**, **420-1**, and **420-2**. For ease of illustration, the individual connections between the gates of the transistors **410-1**, **410-2**, **420-1**, and **420-2** and the blow selector **460** are not shown in FIG. **4**.

[0048] The write circuit **405** also includes transistors **430-1** and **430-2** and transistors **440-1** and **440-2** for selectively blowing the one or more second fuses **260-1** and **260-2**. Each of the transistors **430-1** and **430-2** is coupled between the blow driver **450** and a respective one of the one or more second fuses **260-1** and **260-2**, and each of the transistors **440-1** and **440-2** is coupled between a respective one of the one or more second fuses **260-1** and **260-2** and ground. The gates of the transistors **430-1**, **430-2**, **440-1**, and **440-2** are coupled to the blow selector **460**, which controls the on/off states of the transistors **430-1**, **430-2**, **440-1**, and **440-2**. For ease of illustration, the individual connections between the gates of the transistors **430-1**, **430-2**, **440-1**, and **440-2** and the blow selector **460** are not shown in FIG. **4**.

[0049] In certain aspects, the write circuit **405** is used to program the one or more first fuses **250-1** and **250-2** and the one or more second fuses **260-1** and **260-2** one time at a factory or another facility. In these aspects, the blow driver **450** may be temporarily coupled to an external power source (not shown) at the factory or the other facility to provide the blow driver **450** with power to perform a blow operation. This way, the blow driver **450** is not constrained by the limited power that can be delivered by the energy harvester **120**. After the blow operation, the external power source may be removed. In some implementations, the blow driver **450** may also be external to the device **100**. In these implementations, the blow driver **450** may be temporarily coupled to the device **100** at the factor or the other facility to perform the blow operation.

[0050] To store a bit value of one, the blow selector **460** selects the one or more first fuses **250-1** and **250-2** by turning on the transistors **410-1**, **410-2**, **420-1**, and **420-2** for the one or more first fuses **250-1** and **250-2** while leaving the transistors **430-1**, **430-2**, **440-1**, and **440-2** for the one or more second fuses **260-1** and **260-2** turned off. The blow driver **450** then drives a high current through the one or more first fuses **250-1** and **250-2**. The high current electro-migrates the metal in each of the one or more first fuses **250-1** and **250-2**, which blows the one or more first fuses **250-1** and **250-2** and significantly increases the resistances of the one or more first fuses **250-1** and **250-2**.

[0051] To store a bit value of zero, the blow selector **460** selects the one or more second fuses **260-1** and **260-2** by turning on the transistors **430-1**, **430-2**, **440-1**, and **440-2** for the one or more second fuses **260-1** and **260-2** while leaving the transistors **410-1**, **410-2**, **420-1**, and **420-2** for the one or more first fuses **250-1** and **250-2** turned off. The blow driver **450** then drives a high current through the one or more second fuses **260-1** and **260-2**. The high current electro-migrates the metal in each of the one or more second fuses **260-1** and **260-2**, which blows the one or more second fuses **260-1** and **260-2** and significantly increases the resistances of the one or more second fuses **260-1** and **260-2**.

[0052] FIG. **5** shows an example of a read circuit **510** including the read amplifier **210** according to certain aspects. As discussed further below, the read circuit **510** may be configured to autonomously turn on when the voltage Vdd on a bus **515** reaches a voltage threshold (e.g., during startup) to read the stored bit value. The bus **515** (also referred to as a supply rail) may correspond to the power bus **130** shown in FIG. **1**, which receives power from the energy harvester **120**.

[0053] In this example, the read circuit **510** includes a switch **520** coupled between the bus **515** and the read amplifier **210**. In the example shown in FIG. **5**, the switch **520** is coupled to a power terminal **527** of the read amplifier **210**. The power terminal **527** is coupled to the first terminal **226** of the first inverter **220** (e.g., the source of the first PFET **320**) and the first terminal **236** of the second inverter **230** (e.g., the source of the second PFET **340**).

[0054] The switch **520** is used to selectively gate power to the read amplifier **210**, as discussed further below. The switch **520** has a control input **522** for controlling the on/off state of the switch **520**. As used herein, a “control input” of a switch is an input that controls whether the switch is turned on or turned off based on a signal (e.g., voltage) applied to the input. In the example shown in FIG. **5**, the switch **520** is implemented with a PFET **525** in which the gate of the PFET **525** is coupled to the control input **522**.

[0055] The read circuit **510** also includes a voltage detection circuit **530**, an OR gate **540**, a latch **550**, and a done circuit **560**. The voltage detection circuit **530** has an input **532** coupled to the bus **515**, and an output **534** coupled to the control input **522** of the switch **520** through the OR gate **540**. It is to be appreciated that the output **534** of the voltage detection circuit **530** may be coupled to the control input **522** without the OR gate **540** in some implementations.

[0056] The voltage detection circuit **530** is configured to detect the voltage Vdd on the bus **515** and compare the detected voltage with a voltage threshold. The voltage threshold may be the same as or different from the voltage threshold used by the voltage monitor **145** discussed above. If the detected voltage is below the voltage threshold, then the voltage detection circuit **530** turns off the switch **520**. In this case, the voltage on the bus **515** may be too low for the read amplifier **210** to operate reliably. If the detected voltage is equal to greater than the voltage threshold, then the voltage detection circuit **530** may turn on the switch **520** to power the read amplifier **210**. Thus, in this example, the voltage detection circuit **530** may turn on the switch **520** when the voltage on the bus **515** reaches the voltage threshold. The voltage threshold may be set to a voltage that is sufficient for the read amplifier **210** to reliably read the stored bit value.

[0057] For the example where the switch **520** includes the PFET **525**, the voltage detection circuit **530** turns off the switch **520** by outputting a logic one to the gate of the PFET **525**, and turns on the switch **520** by outputting a logic zero to the gate of the PFET **525**. The logic one may correspond to the voltage Vdd, and the logic zero may correspond to ground potential.

[0058] The latch **550** has a first input **552** coupled to the first output **216** of the read amplifier **210**, and a second input **554** coupled to the second output **218** of the read amplifier **210**. The latch **550** also has a first output **556** and a second output **558**. The latch **550** may be implemented with a set-reset (SR) latch or another type of latch.

[0059] When the switch **520** is turned on, the read amplifier **210** receives power from the bus **515** through the switch **520**. This causes the read amplifier **210** to read the bit value stored by the one or more first fuses **250-1** and **250-2** and the one or more second fuses **260-1** and **260-2**, as discussed above. The read amplifier **210** outputs the read bit value at the first output **216** and the complement of the read bit value at the second output **218**. The latch **550** latches the read bit value and the complement of the read bit value, outputs the latched read bit value at the first output **556**, and outputs the latched complement of the read bit value at the second output **558**. The latched read bit value may be output at a first read output D of the read circuit **510**, and the latched complement of the read bit value may be output at a second read output DB of the read circuit **510**. For example, the first read output D and/or the second read output DB may be coupled to the voltage monitor **145** to provide the read bit value and/or the complement of the read bit value to the voltage monitor **145**.

[0060] The done circuit **560** has a first input **562** coupled to the first output **556** of the latch **550**, and a second input **564** coupled to the second output **558** of the latch **550**. The done circuit **560** also has a first output **566**, a second output **568**, and a third output **570**. The third output **570** may be coupled to the control input **522** of the switch **520** through the OR gate **540**.

[0061] The done circuit **560** is configured to detect when the read operation of the read amplifier **210** is done. For example, the done circuit **560** may detect the read operation is done when one of the outputs **556** and **558** of the latch **550** is high (i.e., has a logic state of one). This is because the outputs **556** and **558** of the latch **550** are complementary after the read operation of the read amplifier **210** is done and the latch **550** latches the read bit value and the complement of the read bit value from the read amplifier **210**. Since the outputs **556** and **558** of the latch **550** are complementary when the read operation is done, one of the outputs **556** and **558** is high when the read operation is done. This allows the done circuit **560** to detect the read operation is done by detecting when one of the outputs **556** and **558** is high.

[0062] When the done circuit **560** detects that the read operation is done, the done circuit **560** outputs a done signal at the first output **566** and outputs a complement of the done signal at the second output **568**. The done signal may be high (e.g., logic one) to indicate that the read operation is done. The done signal may be output at a first done output Done of the read circuit **510**, and the complement of the done signal may be output at a second done output Done_B of the read circuit **510**.

[0063] When the done circuit **560** detects that the read operation is done, the done circuit **560** may also turn off the switch **520** to shut off power to the read amplifier **210**, which reduces current consumption. After the power to the read amplifier **210** is shut off, the latch **550** may maintain the latched read bit value at the first read output D of the read circuit **510** and maintain the latched complement of the read bit value at the second read output DB of the read circuit **510**. For the example where the switch **520** includes the PFET **525**, the done circuit **560** may turn off the switch **520** by outputting a logic one to the gate of the PFET **525** through the OR gate **540**.

[0064] In the example in FIG. **5**, the output **534** of the voltage detection circuit **530** and the third output **570** of the done circuit **560** are coupled to the control input **522** of the switch **520** through the OR gate **540**. In this example, the OR gate **540** has a first input **542** coupled to the output **534** of the voltage detection circuit **530**, a second input **544** coupled to the third output **570** of the done circuit **560**, and an output **546** coupled to the control input **522** of the switch **520**. For the example where the switch **520** includes the PFET **525**, the OR gate **540** turns off the switch **520** when at least one of the outputs **534** and **570** is high, and turns on the switch **520** when both of the outputs **534** and **570** are low. It is to be appreciated that the read circuit **510** is not limited to the OR gate **540**, and that the output **534** of the voltage detection circuit **530** and the third output **570** of the done circuit **560** may be coupled to the control input **522** of the switch **520** through a different type of logic gate or another type of circuit. It is also to be appreciated that the OR gate **540** may be implemented with combinational logic configured to perform an OR operation according to certain aspects.

[0065] FIG. **6** shows an exemplary implementation of the voltage detection circuit **530** according to certain aspects. In this example, voltage detection circuit **530** includes a voltage detector **610** and an inverter **620**. The voltage detector **610** has an input **612** coupled to the input **532** of the voltage detection circuit **530**, and an output **614**. The input of the inverter **620** is coupled to the output **614** of the voltage detector **610**, and the output of the inverter **620** is coupled to the output **534** of the voltage detection circuit **530**.

[0066] The voltage detector **610** is configured to sense the voltage at the input **612**, output a logic zero at the output **614** if the sensed voltage is below the voltage threshold, and output a logic one if the sensed voltage is equal to or greater than the voltage threshold. In the example shown in FIG. **6**, the input **612** of the voltage detector **610** is coupled to the bus **515**, and therefore senses the voltage Vdd on the bus **515** is this example. The voltage detector **610** may be implemented, for example, with a voltage comparator configured to compare the sensed voltage with the voltage threshold, and output a logic one or a logic zero based on the comparison.

[0067] The inverter **620** inverts the output signal of the voltage detector **610** and outputs the inverse of the output signal to the control input **522** of the switch **520** through the OR gate **540**. When the sensed voltage at the input **612** is below the voltage threshold, the voltage detector **610** outputs a logic zero. The inverter **620** inverts the logic zero into a logic one which turns off the switch **520** for the example where the switch **520** includes the PFET **525**. When the sensed voltage at the input **612** is equal to or greater than the voltage threshold, the voltage detector **610** outputs a logic one. The inverter **620** inverts the logic one into a logic zero which turns on the switch **520** for the example where the switch **520** includes the PFET **525**.

[0068] In the example shown in FIG. **6**, the read circuit **510** also includes a gating circuit **630** and an enable circuit **660**. The gating circuit **630** is coupled between the outputs **556** and **558** of the latch **550** and the inputs **562** and **564** of the done circuit **560**. The gating circuit **630** is also coupled to the output **614** of the voltage detector **610**. The enable circuit **660** is coupled between the gating circuit **630** and the read outputs D and DB. As discussed further below, the enable circuit **660** is configured to enable or disable the read outputs D and DB of the read circuit **510**. When the enable circuit **660** enables the read outputs D and DB of the read circuit **510**, the enable circuit **660** couples the gating circuit **630** to the read outputs D and DB.

[0069] The gating circuit **630** is configured to gate the output signals of the latch **550** (i.e., the signals at the outputs **556** and **558**) when the voltage detector **610** detects that the sensed voltage at the input **612** is below the voltage threshold (e.g., the voltage detector **610** outputs a logic zero). The gating circuit **630** is configured to pass the output signals of the latch **550** to the done circuit **560** and the read outputs D and DB of the read circuit **510** when the voltage detector **610** detects that the sensed voltage at the input **612** is equal to or greater than the voltage threshold (e.g., the voltage detector **610** outputs a logic one). In other words, the gating circuit **630** un-gates the outputs **556** and **558** of the latch **550** when the sensed voltage is equal to or greater than the voltage threshold. Thus, in this example, the gating circuit **630** passes the output signals of the latch **550** to the read outputs D and DB of the read circuit **510** when the voltage Vdd on the bus **515** is equal to or greater than the voltage threshold. This helps ensure that a valid read bit and a valid complement of the read bit are output from the first read output D and the second read output DB, respectively.

[0070] In the example shown in FIG. **6**, the gating circuit **630** includes a first AND gate **640** and a second AND gate **650**. The first AND gate **640** has a first input **642** coupled to the first output **556** of the latch **550**, a second **644** input coupled to the output **614** of the voltage detector **610**, and an output **646** coupled to the first input **562** of the done circuit **560**. The second AND gate **650** has a first input **652** coupled to the second output **558** of the latch **550**, a second input **654** coupled to the output **614** of the voltage detector **610**, and an output **656** coupled to the second input **564** of the done circuit **560**. In this example, the first AND gate **640** and the second AND gate **650** gate the signals at the first output **556** and the second output **558**, respectively, of the latch **550** when the output **614** of the voltage detector **610** is low (i.e., logic zero) indicating that the sensed voltage is below the voltage threshold. This first AND gate **640** and the second AND gate **650** pass the signals at the first output **556** and the second output **558**, respectively, of the latch **550** when the output **614** of the voltage detector **610** is high (i.e., logic one) indicating that the sensed voltage is equal to or greater than the voltage threshold.

[0071] It is to be appreciated that the gating circuit **630** is not limited to the AND gates **640** and **650**, and that the gating circuit **630** may be implemented with other types of logic gates in other implementations. It is also to be appreciated that each of the AND gates **640** and **650** may be implemented with combinational logic configured to perform an AND operation according to

certain aspects.
[0072] The enable circuit **660** is configured to enable or disable the read circuit **510** based on an enable signal received at a first enable input EN of the read circuit **510** and a complement of the enable signal received at a second enable input EN_B of the read circuit **510**. In this example, the enable circuit **660** may be configured to disable the read circuit **510** when the enable signal is logic zero and the complement of the enable signal is logic one, and enable the read circuit **510** when the enable signal is logic one and the complement of the enable signal is logic zero.
[0073] In the example shown in FIG. **6**, the second enable input EN_B is coupled to the control input **522** of the switch **520** through the OR gate **540**, which has a third input **690** coupled to the second enable input EN_B in this example. When the complement of the enable signal is logic one to disable the read circuit **510**, the logic one is routed to the control input **522** of the switch **520** through the OR gate **540**. The logic one keeps the switch **520** turned off for the example where the switch **520** includes the PFET **525**. Thus, in this example, the switch **520** is turned off when the read circuit **510** is disabled, which prevents the read amplifier **210** from receiving power from the bus **515**.
[0074] In the example shown in FIG. **6**, the enable circuit **660** includes an AND gate **670** and an OR gate **680**. The AND gate **670** has a first input **672** coupled to the first enable input EN, a second input **674** coupled to the output **656** of the AND gate **650**, and an output **676** coupled to the second read output DB. The OR gate **680** has a first input **682** coupled to the second enable input EN_B, a second input **684** coupled to the output **646** of the AND gate **640**, and an output **686** coupled to the first read output D. When the enable signal and the complement of the enable signal are logic one and logic zero, respectively, the OR gate **680** and the AND gate **670** pass the read bit value and the complement of the read bit value to the first read output D and the second read output DB, respectively.
[0075] It is to be appreciated that the enable circuit **660** is not limited to the AND gate **670** and the OR gate **680**, and that the enable circuit **660** may be implemented with other types of logic gates in other implementations. It is also to be appreciated that AND gate **670** and the OR gate **680** may each be implemented with combinational logic.
[0076] FIG. **7** shows an exemplary implementation of the latch **550** according to certain aspects. In this example, the latch **550** is implemented with an S-R latch including a first NOR gate **710** and a second NOR gate **720**. The first NOR gate **710** has a first input **712** coupled to the first input **552** of the latch **550**, a second input **714**, and an output **716** coupled to the second output **558** of the latch **550**. The second NOR gate **720** has a first input **722** coupled to the second input **554** of the latch **550**, a second input **724** coupled to the output **716** of the first NOR gate **710**, and an output **726** coupled to the second input **714** of the first NOR gate **710** and the first output **556** of the latch **550**. It is to be appreciated that the latch **550** is not limited to the exemplary implementation shown in FIG. **7**. For example, the latch **550** may be implemented with NAND gates or other types of logic gates in other implementations.
[0077] FIG. **8** shows an exemplary implementation of the done circuit **560** according to certain aspects. In this example, the done circuit **560** includes a first voltage detector **810**, a second voltage detector **820**, an OR gate **840**, and a latch **850**. The first voltage detector **810** has an input **812** coupled to the first input **562** of the done circuit **560**, and an output **814**. The second voltage detector **820** has an input **822** coupled to the second input **564** of the done circuit **560**, and an output **824**. The OR gate **840** has a first input **842** coupled the output **814** of the first voltage detector **810**, a second input **844** coupled the output **824** of the second voltage detector **820**, and an output **846** coupled to the latch **850**.
[0078] The first voltage detector **810** is configured to pass a logic one at the input **812** to the output **814** when the logic one is equal to or greater than a pass voltage threshold. The pass voltage threshold may be the same as or different from the voltage threshold of the voltage detector **610**. The second voltage detector **820** is configured to pass a logic one at the input **822** to the output **824**

when the logic one is equal to or greater than the pass voltage threshold. In this example, when the read amplifier **210** is done with a read operation, one of the first and second voltage detectors **810** and **820** receives a logic one at the respective input **812** and **822**. Assuming the logic one is equal to or greater than the pass voltage threshold, the logic one is passed to the first input **842** or the second input **844** of the OR gate **840**. This causes the OR gate **840** to output the logic one to the latch **850**. The latch **850** latches the logic one, and outputs the latched logic one at the first output **566** of the done circuit **560** to provide the done signal, and outputs the complement of the logic one (i.e., logic zero) at the second output **568** of the done circuit **560** to provide the complement of the done signal. As discussed above, the done signal indicates that the read amplifier **210** is done with the read operation.

[0079] In the example shown in FIG. **8**, the latch **850** includes a NOR gate **860** and an inverter **870**. The NOR gate **860** has a first input **862** coupled to the output **846** of the OR gate **840**, a second input **864** coupled to the output of the inverter **870**, and an output **866** coupled to the second output **568** of the done circuit **560** and the input of the inverter **870**. The output of the inverter **870** is coupled to the first output **566** of the done circuit **560**.

[0080] In the example in FIG. **8**, the done signal is used to turn off the switch **520**. For the example where the switch **520** includes the PFET **525**, the logic one of the done signal turns off the switch **520**, shutting off power to the read amplifier **210**.

[0081] As discussed above, the exemplary read circuit **510** may autonomously turn on the read amplifier **210** to read the bit value store at the read circuit **510** when the voltage Vdd on the bus **515** reaches a certain voltage (e.g., the voltage threshold of the voltage detection circuit **530**).

[0082] In certain aspects, a read circuit may turn on to read the respective bit value when another read circuit is done with its read operation. As discussed further below, this feature allows multiple read circuits to be coupled together in a way that causes the read circuits to sequentially read their respective bits values without the need for an external clock signal.

[0083] FIG. **9** shows another example of a read circuit **910** according to certain aspects. As discussed further below, the read circuit **910** is configured to turn on and read its respective bit value when the done signal and/or the complement of the done signal from a previous read circuit (e.g., the read circuit **510** or another instance of the read circuit **910**) indicates the previous read circuit is done with its read operation.

[0084] In this example, the read circuit **910** includes the switch **520**, the voltage detection circuit **530**, the read amplifier **210**, the latch **550**, the done circuit **560**, and the OR gate **540** discussed above. The read circuit **910** also includes a second switch **920** coupled in series with the switch **520** between the bus **515** and the read amplifier **210**. In the description below, the switch **520** is referred to as the first switch. In the example shown in FIG. **9**, the second switch **920** includes a respective PFET **925**. However, it is to be appreciated that the second switch **920** is not limited to this example. Also, in the example shown in FIG. **9**, the input **532** of the voltage detection circuit **530** is coupled between the first switch **520** and the second switch **920**.

[0085] In certain aspects, the second switch **920** is configured to turn on when a previous read circuit (e.g., the read circuit **510** or another instance of the read circuit **910**) indicates that the read operation of the previous read circuit is done. In the example shown in FIG. **9**, the control input **922** of the second switch **920** is coupled to a done input Done_B_in, which is coupled to the second done output Done_B of the previous read circuit to receive the complement of the done signal of the previous read circuit. In this example, the second switch **920** turns on when the complement of the done signal from the previous read circuit is low, indicating that the read operation of the previous read circuit is done. However, it is to be appreciated that the second switch **920** is not limited to this example. In another example, the second switch **920** may be configured to receive the done signal from the previous read circuit and turn on when the done signal from the previous read circuit is high. In both examples, the second switch **920** is configured to turn on when the previous read circuit outputs a signal indicating that the previous read circuit is done with its read operation.

[0086] When the second switch **920** turns on, the first switch **520** and the input **532** of the voltage detection circuit **530** are coupled to the bus **515** through the second switch **920**. This allows the voltage detection circuit **530** to detect the voltage Vdd on the bus **515**, and turn on the first switch **520** when the detected voltage is equal to or greater than the voltage threshold of the voltage detection circuit **530**. This helps ensure that the voltage Vdd on the bus **515** is sufficient to perform a read operation. The bus **515** may be shared with other read circuits (e.g., the previous read circuit).

[0087] When the first switch **520** is turned on, the read amplifier **210** receives power from the bus **515** through the switches **520** and **920**. This causes the read amplifier **210** to read the bit value stored by the respective one or more first fuses **250-1** and **250-2** and the respective one or more second fuses **260-1** and **260-2**, as discussed above. The read amplifier **210** outputs the read bit value at the first output **216** and the complement of the read bit value at the second output **218**. The latch **550** latches the read bit value and the complement of the read bit value, outputs the latched read bit value at the first output **556**, and outputs the latched complement of the read bit value at the second output **558**. The latched read bit value may be output at the first read output D of the read circuit **910**, and the latched complement of the read bit value may be output at the second read output DB of the read circuit **910**. The done circuit **560** detects that the read operation is done based on the outputs **556** and **558** of the latch, and outputs the done signal at the first done output Done and outputs the complement of the done signal at the second done output Done_B indicating that the read operation is done. For example, the done signal and/or the complement of the done signal may be coupled to a next read circuit (e.g., another instance of the read circuit **910**) to trigger the next read circuit to read its respective bit value.

[0088] In certain aspects, the done circuit **560** may turn off both the first switch **520** and the second switch **920** when the done circuit **560** detects that the read operation is done. In this example, the third output **570** of the done circuit is coupled to the control input **522** of the first switch **520** through the OR gate **540**, as discussed above. Also, in this example, the third output **570** is coupled to the control input **922** of the second switch **920** through a second OR gate **930**. In this example, the done circuit **560** may turn off both switches **520** and **920** by outputting a logic one at the third output **570**, in which the logic one passes through the OR gate **540** to turn off the first switch **520** and passes through the second OR gate **930** to turn off the second switch **920**. In the example shown in FIG. **9**, the second OR gate **930** has a first input **932** coupled to the done input Done_B_in, a second input **934** coupled to the third output **570** of the done circuit **560**, and an output **936** coupled to the control input **922** of the second switch **920**.

[0089] FIG. **10** shows an example in which the read circuit **910** also includes the gating circuit **630** and the enable circuit **660** discussed above. In this example, the gating circuit **630** includes the AND gates **640** and **650** in which each of the AND gates **640** and **650** has a third input coupled to a done input Done_in, which may be coupled to the first done output Done of the previous read circuit to receive the done signal of the previous read circuit. In this example, the gating circuit **630** is ungated when the voltage Vdd on the bus **515** is equal to or greater than the voltage threshold of the voltage detection circuit **530** and the done signal from the previous read circuit indicates that the read operation of the previous read circuit is done.

[0090] The latch **550** may be implemented with the example implementation shown in FIG. **7**, and the done circuit **560** may be implemented with the exemplary implementation shown in FIG. **8**. However, it is to be appreciated that the latch **550** and the done circuit **560** are not limited to these examples.

[0091] FIG. **11** shows an example of multiple read circuits coupled together to read multiple bits. The multiple read circuits include the read circuit **510** and read circuits **910-1** to **910-3** where each of the read circuits **910-1** to **910-3** is a separated instance (i.e., copy) of the read circuit **910**. The read circuit **510** and the read circuits **910-1** to **910-3** may share the bus **515**.

[0092] In this example, the first done output Done and the second done output Done_B of the read circuit **510** are coupled to the done input Done_in and the done input Done_B_in of the read circuit **910-1**, respectively. The first done output Done and the second done output Done_B of the read circuit **910-1** are coupled to the done input Done_in and the done input Done_B_in of the read circuit **910-2**, respectively. The first done output Done and the second done output Done_B of the read circuit **910-2** are coupled to the done input Done_in and the done input Done_B_in of the read circuit **910-3**, respectively. The first done output Done and the second done output Done_B of the read circuit **910-3** may be coupled to the done input Done_in and the done input Done_B_in of a next read circuit (not shown), respectively.
[0093] During operation, when the voltage vdd on the bus **515** reaches a certain voltage (e.g., the voltage threshold of the voltage detection circuit **530** in the read circuit **510**), the read circuit **510** turns on and read its respective bit. When the read operation of the read circuit **510** is done, the read circuit **510** outputs its read bit (labeled "Bit**1**") and outputs the respective done signal and the complement of the respective done signal to the read circuit **910-1**. This triggers the read circuit **910-1** to turn on and read its respective bit. When the read operation of the read circuit **910-1** is done, the read circuit **910-1** outputs its read bit (labeled "Bit**2**") and outputs the respective done signal and the complement of the respective done signal to the read circuit **910-2**. This triggers the read circuit **910-2** to turn on and read its respective bit. When the read operation of the read circuit **910-2** is done, the read circuit **910-2** outputs its read bit (labeled "Bit**3**") and outputs the respective done signal and the complement of the respective done signal to the read circuit **910-3**. This triggers the read circuit **910-3** to turn on and read its respective bit, and so forth.
[0094] Thus, in this example, the read circuit **510** and the read circuit **910-1** to **910-3** sequentially read and output their respective bits, in which each read circuit triggers the next read circuit to read its bit until all of the read circuits have been triggered. The read circuit **510** and the read circuits **910-1** to **910-3** are able to sequentially read and output their respective bit values without the need for an external clock signal to coordinate the read operations of the read circuit **510** and the read circuits **910-1** to **910-3**. The removal of the clock signal reduces power consumption. Also, the sequential reading of the bits and the voltage detection circuit (e.g., respective voltage detection circuit **530**) in each of the read circuit **510** and the read circuits **910-1** to **910-3** helps ensure that the voltage Vdd on the shared bus **515** is high enough for each read operation.
[0095] FIG. **12** shows another example in which the read circuit **510** and read circuits **910-1** to **910-3** are coupled together. In this example, the first read output D and the second read output DB of the read circuit **510** are coupled to the first enable input EN and the second enable input EN_B, respectively, of each of the read circuits **910-1** to **910-3**. As a result, the read circuits **910-1** to **910-3** are enabled or disabled based on the read bit of the read circuit **510**. In this example, the read circuits **910-1** to **910-3** are enabled when the read bit of the read circuit **510** has a bit value of one, and disabled when the read bit of the read circuit **510** has a bit value of zero. Thus, in this example, only the bit of the read circuit **510** (labeled "Bit**1**") is read when the bit of the read circuit **510** has a bit value of zero.
[0096] As discussed above, the non-volatile memory **135** (which includes the read circuit **510** and the read circuits **910-1** to **910-3** in this example) may store the voltage threshold for the voltage monitor **145**. In some implementations, the voltage threshold may be a function of the process corner of the circuit **150**. For example, the voltage threshold may be higher for a slow-slow (SS) process corner and lower for a fast-fast (FF) process corner.
[0097] In this example, the voltage monitor **145** may be configured with a default voltage threshold for the SS process corner, and the bit of the read circuit **510** may indicate whether the process corner for the circuit **150** is an SS process corner. In this example, a bit value of zero indicates the SS process corner. Thus, when the process corner for the circuit **150** is an SS process corner, the read circuit **510** outputs a bit value of zero to the voltage monitor **145** indicating the SS process corner, and the voltage monitor **145** uses the default voltage threshold. In this case, the remaining

read circuits **910-1** to **910-3** are disabled to reduce power. When the process corer for the circuit **150** is another process corner (e.g., FF process corner or typical-typical (TT) process corner), the read circuit **510** outputs a bit value of one and enables the remaining read circuits **910-1** to **910-3** to read and output their respective bits to the voltage monitor **145**. The bits of the remaining read circuits **910-1** to **910-3** may indicate a voltage threshold for the voltage monitor **145**.

[0098] FIG. **13** shows an example of a method **1300** for reading a bit stored in electronic fuses. The electronic fuses include one or more first fuses (e.g., the one or more first fuses **250-1** and **250-2**) and one or more second fuses (e.g., the one or more second fuses **260-1** and **260-2**).

[0099] At block **1310**, a resistance difference between the one or more first fuses and the one or more second fuses is sensed using a read amplifier, the read amplifier including a first inverter and a second inverter that are cross coupled. For example, the read amplifier may correspond to the read amplifier **210**, the first inverter may correspond to the first inverter **220**, and the second inverter may correspond to the second inverter **230**. In one example, each of the one or more first fuses is blown and each of the one or more second fuses is unblown. In another example, each of the one or more first fuses is unblown and each of the one or more second fuses is blown.

[0100] At block **1320**, a logic value at an output of the first inverter is latched to obtain a first latched logic value. For example, the logic value at the output of the first inverter may be latched by the latch **550** (e.g., an S-R latch).

[0101] At block **1330**, a logic value at an output of the second inverter is latched to obtain a second latched logic value, wherein the first latched logic value and the second latched logic value are complementary. For example, the logic value at the output of the second inverter may be latched by the latch **550** (e.g., an S-R latch).

[0102] At block **1340**, at least one of the first latched logic value and the second latched logic value is output. For example, the at least one of the first latched logic value and the second latched logic value may be output from at least one of the first read output D and the second read output DB.

[0103] In certain aspects, the method **1300** may further include detecting one of the first latched logic value and the second latched logic value is a logic one, and outputting a done signal in response to detecting the one of the first latched logic value and the second latched logic value is the logic one. For example, the done circuit **560** may detect one of the first latched logic value and the second latched logic value is a logic one, and output the done signal.

[0104] In certain aspects, a switch (e.g., switch **520**) is coupled between a bus (e.g., the bus **515**) and the read amplifier. In these aspects, the method **1300** may further include detecting a voltage on the bus is above a threshold, and turning on the switch is response to detecting the voltage on the bus is above the threshold. For example, the voltage detection circuit **530** may detect the voltage on the bus and turn on the switch. The method **1300** may further include detecting one of the first latched logic value and the second latched logic value is a logic one, and turning off the switch in response to detecting the one of the first latched logic value and the second latched logic value is the logic one. For example, the done circuit **560** may detect one of the first latched logic value and the second latched logic value is a logic one, and turn off the switch in response to detecting the one of the first latched logic value and the second latched logic value is the logic one.

[0105] In certain aspects, a first switch (e.g., switch **520**) and a second switch (e.g., switch **920**) are coupled in series between a bus (e.g., bus **515**) and the read amplifier. In these aspects, the method **1300** may further include receiving a signal indicating a read operation of a read circuit for another bit is done, turning on the second switch in response to receiving the signal, detecting a voltage between the first switch and the second switch is above a threshold, and turning on the first switch is response to detecting the voltage between the first switch and the second switch is above the threshold. For example, the voltage detection circuit **530** may detect the voltage between the first switch and the second switch and turn on the first switch. The method **1300** may further include detecting one of the first latched logic value and the second latched logic value is a logic one, and turning off the first switch and the second switch in response to detecting the one of the first latched logic value and the second latched logic value is the logic one. For example, the done circuit **560** may detect one of the first latched logic value and the second latched logic value is a logic one, and turn off the first switch and the second switch in response to detecting the one of the first latched logic value and the second latched logic value is the logic one.

[0106] Implementation examples are described in the following numbered clauses: [0107] 1. A system, comprising: [0108] a read amplifier having a first input, a second input, a first output, and a second output, the read amplifier comprising: [0109] a first inverter having an input and an output, wherein the output of the first inverter is coupled to the first output of the read amplifier; and [0110] a second inverter having an input and an output, wherein the output of the second inverter is coupled to the second output of the read amplifier and the input of the first inverter, and the input of the second inverter is coupled to the output of the first inverter; [0111] one or more first fuses coupled to the first input of the read amplifier; and [0112] one or more second fuses coupled to the second input of the read amplifier. [0113] 2. The system of clause 1, wherein each of the one or more first fuses is blown and each of the one or more second fuses is unblown. [0114] 3. The system of clause 1, wherein each of the one or more first fuses is unblown and each of the one or more second fuses is blown. [0115] 4. The system of any one of clauses 1 to 3, wherein the one or more first fuses are coupled in series between the first input of the read amplifier a ground, and the one or more second fuses are coupled in series between the second input of the read amplifier and the ground. [0116] 5. The system of any one of clauses 1 to 4, wherein the first inverter comprises: [0117] a first n-type field effect transistor (NFET), wherein a source of the first NFET is coupled to the first input of the read amplifier, a drain of the first NFET is coupled to the output of the first inverter, and a gate of the first NFET is coupled to the input of the first inverter; and [0118] a first p-type field effect transistor (PFET), wherein a source of the first PFET is coupled to a power terminal of the read amplifier, a drain of the first PFET is coupled to the output of the first inverter, and a gate of the first PFET is coupled to the input of the first inverter. [0119] 6. The system of clause 5, wherein the second inverter comprises: [0120] a second NFET, wherein a source of the second NFET is coupled to the second input of the read amplifier, a drain of the second NFET is coupled to the output of the second inverter, and a gate of the second NFET is coupled to the input of the second inverter; and [0121] a second PFET, wherein a source of the second PFET is coupled to the power terminal of the read amplifier, a drain of the second PFET is coupled to the output of the second inverter, and a gate of the second PFET is coupled to the input of the second inverter. [0122] 7. The system of any one of clauses 1 to 6, further comprising: [0123] a switch coupled between a bus and the read amplifier; and [0124] a voltage detection circuit having an input and an output, wherein the input of the voltage detection circuit is coupled to the bus, and the output of the voltage detection circuit is coupled to a control input of the switch. [0125] 8. The system of clause 7, wherein the voltage detection circuit is configured to: [0126] detect a voltage at the input of the voltage detection circuit; [0127] turn off the switch if the detected voltage is less than a voltage threshold; and [0128] turn on the switch if the detected voltage is greater than the voltage threshold. [0129] 9. The system of clause 7 or 8, further comprising an energy harvester coupled to the bus. [0130] 10. The system of any one of clauses 7 to 9, further comprising a latch having a first input, a second input, a first output, and a second output, wherein the first input of the latch is coupled to the first output of the read amplifier, and the second input of the latch is coupled to the second output of the read amplifier. [0131] 11. The system of clause 10, wherein the latch comprises a set-reset (S-R) latch. [0132] 12. The system of clause 10 or 11, further comprising a done circuit coupled to the first output of the latch and the second output of the latch, wherein the done circuit is configured to: [0133] detect one of the first output of the latch and the second output of the latch has a logic state of one; and [0134] output a done signal when the done circuit detects the one of the first output of the latch and the second output of the latch has the logic state of one. [0135] 13. The system of clause 12, wherein the done circuit is configured to turn off the switch when the done circuit detects the one of the first output of the latch and the second output of the latch has the logic state of one. [0136] 14. The system of any one of clauses 10 to 13, further comprising a gating circuit, wherein the gating circuit is coupled between the first output of the latch and a first read output, and the gating circuit is coupled between the second output of the latch and a second read output. [0137] 15. The system of clause 14, wherein the voltage detection circuit is configured to: [0138] detect a voltage at the input of the voltage detection circuit; [0139] cause the gating circuit to gate the first output of the latch and the second output of the latch if the detected voltage is less than a voltage threshold; and [0140] cause the gating circuit to un-gate the first output of the latch and the second output of the latch if the detected voltage is greater than the voltage threshold. [0141] 16. The system of clause 15, wherein the voltage detection circuit is configured to: [0142] turn off the switch if the detected voltage is less than a voltage threshold; and [0143] turn on the switch if the detected voltage is greater than the voltage threshold. [0144] 17. The system of any one of clauses 1 to 6, further comprising: [0145] a first switch; [0146] a second switch, wherein the first switch and the second switch are coupled in series between a bus and the read amplifier; and [0147] a voltage detection circuit having an input and an output, wherein the input of the voltage detection circuit is coupled between the first switch and the second switch, and the output of the voltage detection circuit is coupled to a control input of the first switch. [0148] 18. The system of clause 17, wherein a control input of the second switch is configured to receive a signal from a read circuit indicating the read circuit is done with a read operation, and the second switch is configured to turn on in response to the signal from the read circuit. [0149] 19. The system of clause 17 or 18, wherein the voltage detection circuit is configured to: [0150] detect a voltage at the input of the voltage detection circuit; [0151] turn off the first switch if the detected voltage is less than a voltage threshold; and [0152] turn on the first switch if the detected voltage is greater than the voltage threshold. [0153] 20. The system of any one of clauses 1 to 19, wherein: [0154] the first inverter has a first terminal and a second terminal; [0155] the second inverter has a first terminal and a second terminal; [0156] the first terminal of the first inverter and the first terminal of the second inverter are coupled to a power terminal of the read amplifier; [0157] the second terminal of the first inverter is coupled to the first input of the read amplifier; and [0158] the second terminal of the second inverter is coupled to the second input of the read amplifier. [0159] 21. The system of clause 20, further comprising: [0160] a switch coupled between a bus and the power terminal of the read amplifier; and [0161] a voltage detection circuit having an input and an output, wherein the input of the voltage detection circuit is coupled to the bus, and the output of the voltage detection circuit is coupled to a control input of the switch. [0162] 22. The system of clause 21, wherein the voltage detection circuit is configured to: [0163] detect a voltage at the input of the voltage detection circuit; [0164] turn off the switch if the detected voltage is less than a voltage threshold; and [0165] turn on the switch if the detected voltage is greater than the voltage threshold. [0166] 23. A method for reading a bit stored in electronic fuses, the electronic fuses comprising one or more first fuses and one or more second fuses, the method comprising: [0167] sensing a resistance difference between the one or more first fuses and the one or more second fuses using a read amplifier, the read amplifier including a first inverter and a second inverter that are cross coupled; [0168] latching a logic value at an output of the first inverter to obtain a first latched logic value; [0169] latching a logic value at an output of the second inverter to obtain a second latched logic value, wherein the first latched logic value and the second latched logic value are complementary; and [0170] outputting at least one of the first latched logic value and the second latched logic value. [0171] 24. The method of clause 23, further comprising: [0172] detecting one of the first latched logic value and the second latched logic value is a logic one; and [0173] outputting a done signal in response to detecting the one of the first latched logic value and the second latched logic value is the logic one. [0174] 25. The method of clause 24, further comprising outputting the done signal to a read circuit configured to read another bit. [0175] 26. The method of any one of clauses 23 to 25, wherein a switch is coupled between a bus and the read amplifier, and the method further comprising: [0176] detecting a voltage on the bus is above a threshold; and [0177] turning on the switch is response to detecting the voltage on the bus is above the threshold. [0178] 27. The method of clause 26, further comprising: [0179] detecting one of the first latched logic value and the second latched logic value is a logic one; and [0180] turning off the switch in response to detecting the one of the first latched logic value and the second latched logic value is the logic one. [0181] 28. The method of any one of clauses 23 to 25, wherein a first switch and a second switch are coupled in series between a bus and the read amplifier, and the method further comprises: [0182] receiving a signal indicating a read operation of a read circuit for another bit is done; [0183] turning on the second switch in response to receiving the signal; [0184] detecting a voltage between the first switch and the second switch is above a threshold; and [0185] turning on the first switch is response to detecting the voltage between the first switch and the second switch is above the threshold. [0186] 29. The method of clause 28, further comprising: [0187] detecting one of the first latched logic value and the second latched logic value is a logic one; and [0188] turning off the first switch and the second switch in response to detecting the one of the first latched logic value and the second latched logic value is the logic one. [0189] 30. The method of any one of clauses 23 to 29, wherein each of the one or more first fuses is blown and each of the one or more second fuses is unblown. [0190] 31. The method of any one of clauses 23 to 29, wherein each of the one or more first fuses is unblown and each of the one or more second fuses is blown.

[0191] Within the present disclosure, the word “exemplary” is used to mean “serving as an example, instance, or illustration.” Any implementation or aspect described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects of the disclosure. Likewise, the term “aspects” does not require that all aspects of the disclosure include the discussed feature, advantage or mode of operation. The term “coupled” is used herein to refer to the direct or indirect electrical coupling between two structures. It is also to be appreciated that the term “ground” may refer to a DC ground or an AC ground, and thus the term “ground” covers both possibilities.

[0192] Any reference to an element herein using a designation such as “first,” “second,” and so forth does not generally limit the quantity or order of those elements. Rather, these designations are used herein as a convenient way of distinguishing between two or more elements or instances of an element. Thus, a reference to first and second elements does not mean that only two elements can be employed, or that the first element must precede the second element.

[0193] The previous description of the disclosure is provided to enable any person skilled in the art to make or use the disclosure. Various modifications to the disclosure will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other variations without departing from the spirit or scope of the disclosure. Thus, the disclosure is not intended to be limited to the examples described herein but is to be accorded the widest scope consistent with the principles and novel features disclosed herein.

Claims

1. A system, comprising: a read amplifier having a first input, a second input, a first output, and a second output, the read amplifier comprising: a first inverter having an input and an output, wherein the output of the first inverter is coupled to the first output of the read amplifier; and a second inverter having an input and an output, wherein the output of the second inverter is coupled to the second output of the read amplifier and the input of the first inverter, and the input of the second inverter is coupled to the output of the first inverter; one or more first fuses coupled to the first input of the read amplifier; and one or more second fuses coupled to the second input of the read amplifier.

2. The system of claim 1, wherein each of the one or more first fuses is blown and each of the one or more second fuses is unblown.

3. The system of claim 1, wherein each of the one or more first fuses is unblown and each of the one or more second fuses is blown.
4. The system of claim 1, wherein the first inverter comprises: a first n-type field effect transistor (NFET), wherein a source of the first NFET is coupled to the first input of the read amplifier, a drain of the first NFET is coupled to the output of the first inverter, and a gate of the first NFET is coupled to the input of the first inverter; and a first p-type field effect transistor (PFET), wherein a source of the first PFET is coupled to a power terminal of the read amplifier, a drain of the first PFET is coupled to the output of the first inverter, and a gate of the first PFET is coupled to the input of the first inverter.
5. The system of claim 4, wherein the second inverter comprises: a second NFET, wherein a source of the second NFET is coupled to the second input of the read amplifier, a drain of the second NFET is coupled to the output of the second inverter, and a gate of the second NFET is coupled to the input of the second inverter; and a second PFET, wherein a source of the second PFET is coupled to the power terminal of the read amplifier, a drain of the second PFET is coupled to the output of the second inverter, and a gate of the second PFET is coupled to the input of the second inverter.
6. The system of claim 1, further comprising: a switch coupled between a bus and the read amplifier; and a voltage detection circuit having an input and an output, wherein the input of the voltage detection circuit is coupled to the bus, and the output of the voltage detection circuit is coupled to a control input of the switch.
7. The system of claim 6, wherein the voltage detection circuit is configured to: detect a voltage at the input of the voltage detection circuit; turn off the switch if the detected voltage is less than a voltage threshold; and turn on the switch if the detected voltage is greater than the voltage threshold.
8. The system of claim 6, further comprising a latch having a first input, a second input, a first output, and a second output, wherein the first input of the latch is coupled to the first output of the read amplifier, and the second input of the latch is coupled to the second output of the read amplifier.
9. The system of claim 8, further comprising a done circuit coupled to the first output of the latch and the second output of the latch, wherein the done circuit is configured to: detect one of the first output of the latch and the second output of the latch has a logic state of one; and output a done signal when the done circuit detects the one of the first output of the latch and the second output of the latch has the logic state of one.
10. The system of claim 9, wherein the done circuit is configured to turn off the switch when the done circuit detects the one of the first output of the latch and the second output of the latch has the logic state of one.
11. The system of claim 1, further comprising: a first switch; a second switch, wherein the first switch and the second switch are coupled in series between a bus and the read amplifier; and a voltage detection circuit having an input and an output, wherein the input of the voltage detection circuit is coupled between the first switch and the second switch, and the output of the voltage detection circuit is coupled to a control input of the first switch.
12. The system of claim 11, wherein a control input of the second switch is configured to receive a signal from a read circuit indicating the read circuit is done with a read operation, and the second switch is configured to turn on in response to the signal from the read circuit.
13. The system of claim 12, wherein the voltage detection circuit is configured to: detect a voltage at the input of the voltage detection circuit; turn off the first switch if the detected voltage is less than a voltage threshold; and turn on the first switch if the detected voltage is greater than the voltage threshold.
14. The system of claim 1, wherein: the first inverter has a first terminal and a second terminal; the second inverter has a first terminal and a second terminal; the first terminal of the first inverter and

the first terminal of the second inverter are coupled to a power terminal of the read amplifier; the second terminal of the first inverter is coupled to the first input of the read amplifier; and the second terminal of the second inverter is coupled to the second input of the read amplifier.

15. The system of claim 14, further comprising: a switch coupled between a bus and the power terminal of the read amplifier; and a voltage detection circuit having an input and an output, wherein the input of the voltage detection circuit is coupled to the bus, and the output of the voltage detection circuit is coupled to a control input of the switch.

16. The system of claim 15, wherein the voltage detection circuit is configured to: detect a voltage at the input of the voltage detection circuit; turn off the switch if the detected voltage is less than a voltage threshold; and turn on the switch if the detected voltage is greater than the voltage threshold.

17. A method for reading a bit stored in electronic fuses, the electronic fuses comprising one or more first fuses and one or more second fuses, the method comprising: sensing a resistance difference between the one or more first fuses and the one or more second fuses using a read amplifier, the read amplifier including a first inverter and a second inverter that are cross coupled; latching a logic value at an output of the first inverter to obtain a first latched logic value; latching a logic value at an output of the second inverter to obtain a second latched logic value, wherein the first latched logic value and the second latched logic value are complementary; and outputting at least one of the first latched logic value and the second latched logic value.

18. The method of claim 17, further comprising: detecting one of the first latched logic value and the second latched logic value is a logic one; and outputting a done signal in response to detecting the one of the first latched logic value and the second latched logic value is the logic one.

19. The method of claim 17, wherein a switch is coupled between a bus and the read amplifier, and the method further comprising: detecting a voltage on the bus is above a threshold; and turning on the switch is response to detecting the voltage on the bus is above the threshold.

20. The method of claim 17, wherein a first switch and a second switch are coupled in series between a bus and the read amplifier, and the method further comprises: receiving a signal indicating a read operation of a read circuit for another bit is done; turning on the second switch in response to receiving the signal; detecting a voltage between the first switch and the second switch is above a threshold; and turning on the first switch is response to detecting the voltage between the first switch and the second switch is above the threshold.